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ANALYSIS OF A PASSIVE DECAY HEAT REMOVAL SYSTEM

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Contents

Introduction	2
1 Problem Statement	3
2 Heat Exchanger	4
2.1 Heat Exchanger 1 (HX1):	5
2.2 Heat Exchanger 2 (HX2):	6
3 Power Balance	6
4 Intermediate System Circuit	7
4.1 Energy Balance in Heat Exchanger 2	7
4.2 Momentum Balance	9
5 Primary System Circuit (PSC)	11
5.1 Thermal Coupling with HX1	11
5.2 Momentum Balance and Core Integration	11
5.3 Vessel and Loop Pressure Drops	12
6 Numerical Methodology	12
6.1 Iterative Scheme	12
6.2 Convergence and Relaxation	14
7 Results and Discussion	14
7.1 Intermediate System Circuit Results	14
7.2 Primary System Circuit Results	15
7.3 Physical Interpretation of the Results	16
Conclusions	17
References	18

Introduction

The decay of fission products generates a non-negligible thermal power even after reactor shutdown, commonly referred to as *decay heat*. The continuous removal of this residual heat is essential to prevent temperature escalation and potential damage to the reactor core. For this purpose, nuclear power plants are equipped with a Decay Heat Removal System (DHRS), which can operate according to either active or passive principles.

In most currently operating Pressurized Water Reactors (PWRs), an active DHRS is adopted. In such systems, forced circulation is ensured by pumps that drive the primary coolant through the reactor core, where residual thermal power is extracted. The heated coolant is subsequently routed to a heat exchanger, where heat is transferred to a secondary fluid or thermal sink. In the event of a loss of external electrical power, emergency diesel generators are typically employed to guarantee continued pump operation and thus maintain adequate heat removal.

Conversely, passive DHRS configurations provide enhanced safety characteristics by eliminating the need for external power sources or mechanical drivers. These systems rely on natural circulation mechanisms, driven by density gradients within the fluid. Typically, in-pool heat exchangers are utilized as ultimate heat sinks, allowing heat to be dissipated without active intervention.

From a system-level perspective, the DHRS can be described as comprising three interconnected cooling circuits. The Primary System Circuit (PSC) is responsible for removing heat directly from the reactor core. The Intermediate System Circuit (ISC) receives thermal energy from the primary side via a first heat exchanger (HX1). Finally, the Tertiary System Circuit (TSC) transfers heat from a second heat exchanger (HX2) to the environment, often through steam generation and subsequent condensation in a cooling tower. Within this framework, the reactor core represents the heat source, while HX1, HX2, and the cooling tower serve as thermal coupling elements between circuits, and the atmosphere acts as the ultimate heat sink.

Under normal operating conditions, the DHRS remains inactive. A check valve located in the cold leg of the primary circuit prevents coolant circulation within the system. However, in accidental scenarios—such as the failure of primary coolant pumps—the resulting pressure differential across the valve induces its opening, thereby initiating the DHRS operation.

The activation of passive circulation is governed by density differences between the colder fluid in the reactor vessel downcomer and the hotter fluid within the core. This buoyancy-driven mechanism establishes natural circulation in the PSC, progressively heating the hot leg and the primary side of HX1. Consequently, heat transfer to the ISC initiates circulation within the intermediate loop, leading to the heating of the pool side of HX2 until saturation conditions are achieved. The generated steam is then conveyed through the TSC to a cooling tower, where it is condensed and heat is ultimately rejected to the atmosphere.

The present analysis focuses on the steady-state behavior of the DHRS following its activation and the attainment of saturation conditions on the pool side of HX2. At this stage, the residual thermal power to be removed corresponds to approximately 1% of the nominal reactor thermal power. The primary quantities of interest include the mass flow rates within the PSC and ISC, as well as the temperatures in the hot and cold legs of both circuits.

1 Problem Statement

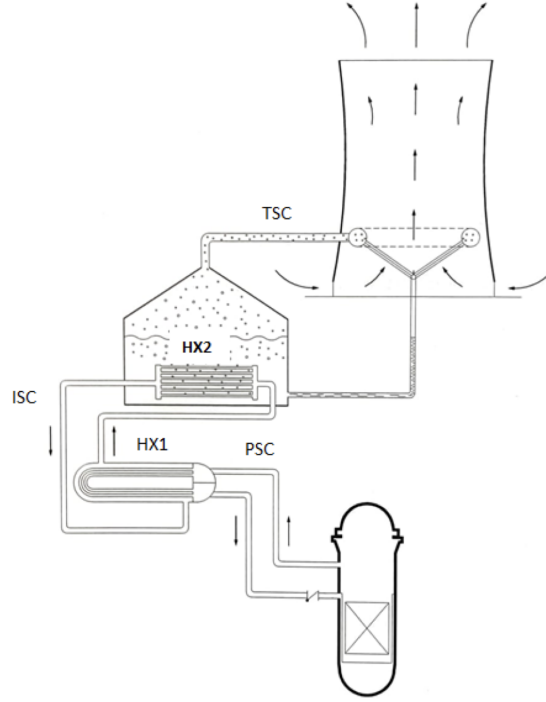


Figure 1: Passive Decay Heat Removal System of a nuclear fission reactor

Assumptions:

- System is in steady state, the initial transient following reactor shutdown is neglected¹
- The cooling tower exhausts all the power necessary to keep the pool at saturation conditions
- Constant coolant pressure in the three circuits to compute water and steam properties
- No fouling resistance in the heat exchangers
- HX1 not pure countercurrent (correction factor 0.7)

Data:

- Nominal thermal power: $P_{nom} = 600 \text{ MW}_{th}$
- $\Delta p_{vessel} = 1.2 \text{ bar}$ for $\dot{m} = 3200 \text{ kg/s}^2$
- Primary System Circuit (PSC) data:

¹Immediately after reactor shutdown the core thermal power drops to approximately 7% of its nominal value. The power then reaches a quasy-steady state at 1% of the nominal thermal power after few hours.

²A constant loss coefficient with the flow rate It is assumed - not a constant pressure drop.

Data	Symbol	Value	Units
Coolant pressure	p	75	bar
Outer pipe diameter	D_{out}	40.64×10^{-2}	m
Pipe wall thickness	t_{pipe}	1.031×10^{-3}	m
Pipe length	L	16	m
Pipe relative roughness	ε/D	$2 \cdot 10^{-4}$	-
Pressure loss coefficient for 90° bends	k_{loss}	0.45	-
Number of 90° bends	N_{bends}	4	-
Valve pressure loss coefficient	$k_{\text{loss, valve}}$	0.12	-
Elevation between hot and cold leg	H_1	7	m
Elevation between top and bottom of the core	H_2	3	m

- Heat Exchanger 1 (HX1) data:

Data	Symbol	Value	Units
Outer tube diameter	$D_{\text{out, tube}}$	19.05	mm
Tube thickness	t	1.24	mm
Number of tubes	N_t	897	-
Pitch	d_p	28.5	mm
Shell inner diameter	D_{shell}	1.5	m
Number of baffles	N_b	2	-
Baffles spacing	l_{baffles}	1.6	m
Average tube length	L_{tubes}	9.314	m
Tube thermal conductivity	k_{th}	15	$W/(mK)$
Headers average flow area	A_h	0.883	m^2
Heat transfer area	$A_{\text{ht, HX1}}$	500	m^2
Tube relative roughness	ε/D	10^{-4}	-
Correction factor for ΔT_{ml}	F_T	0.7	-

- Intermediate System Circuit (ISC) data:

Data	Symbol	Value	Units
Coolant pressure	p	70	bar
Outer tube diameter	$D_{\text{out, tube}}$	40.64×10^{-2}	m
Pipe wall thickness	t_{pipe}	1.031×10^{-3}	m
Total pipe length	L_{pipe}	40	m
Pressure loss coefficient for 90° bends	k_{loss}	0.45	-
Number of 90° bends	N_{bends}	2	-
Tube relative roughness	ε/D	$2 \cdot 10^{-4}$	-
Net elevation	H_{ISC}	10	m

- Heat Exchanger 2 (HX2) data:

Data	Symbol	Value	Units
Tube outer diameter	$D_{\text{out, tube}}$	25.4	mm
Tube thickness	t	1.24	mm
Number of tubes	N_t	770	-
Average tube length	L_{tubes}	7	m
Manifold diameter	D_{manifold}	16	inches
Tube thermal conductivity	k_{th}	15	$W m^{-1} K^{-1}$
Heat transfer area	$A_{\text{ht, HX2}}$	430	m^2
Tube relative roughness	ε/D	10^{-4}	-

2 Heat Exchanger

Before proceeding with the solution of the problem, it is necessary to carry out a preliminary analysis of the heat exchangers HX1 and HX2 in order to determine the main geometrical and fluid-dynamic parameters that will be employed in the subsequent calculations.

In the heat exchanger industry, including applications in conventional and nuclear power plants, two main tube bundle configurations are commonly adopted: *straight-tube* and *U-tube* heat exchangers. The schematic provided in the problem statement clearly shows that both configurations are employed within the considered system.

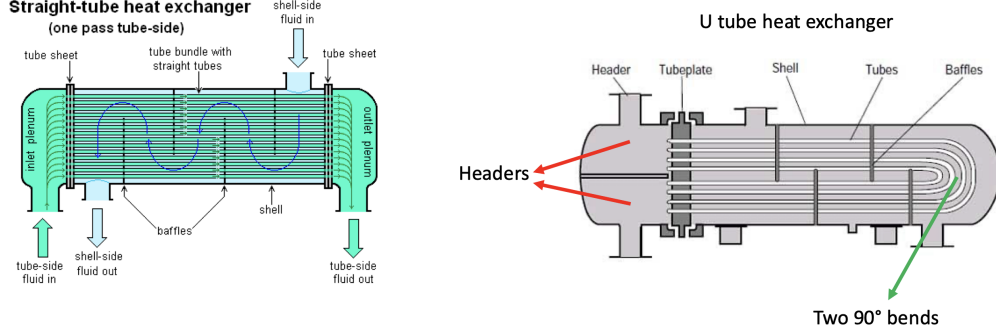


Figure 2: Sample of straight-tubes heat exchanger (right)[2] and U-shaped tubes (left)[2].

2.1 Heat Exchanger 1 (HX1):

HX1 is a counter-current shell-and-tube heat exchanger equipped with U-tubes arranged in a triangular lattice configuration. The main technical and geometrical data associated with this component are reported in Table 2.

The primary function of a heat exchanger is, naturally, the transfer of thermal energy between two fluids. The heat transfer coefficient on the tube side can be evaluated through standard correlations available in the literature. Conversely, for the shell side, a suitable empirical correlation for the evaluation of the convective heat transfer coefficient is given by:

$$h = 0.351 \frac{k_{th}}{D_{eq,shell}} Re_{shell}^{0.55} Pr^{1/3} \quad (1)$$

where k_{th} is the thermal conductivity of the fluid, Pr is the Prandtl number, Re is the Reynolds number, and $D_{eq,shell}$ is the equivalent shell-side diameter. For a triangular tube arrangement, the latter can be expressed as:

$$D_{eq,shell} = \frac{2\sqrt{3}d_p^2}{\pi D_{tube}} - D_{tube} = 0.028 \text{ m} \quad (2)$$

where d_p is the tube pitch and D_{tube} is the outer diameter of the heat exchanger tubes.

The shell-side flow area can instead be computed as:

$$A_{shell} = \frac{D_{shell}}{d_p} (d_p - D_{tube}) l_{baffles} = 0.80 \text{ m}^2 \quad (3)$$

where D_{shell} is the inner shell diameter and $l_{baffles}$ is the spacing between adjacent baffles. Once the shell-side flow area is known, the corresponding Reynolds number can be evaluated, allowing the determination of the shell-side heat transfer coefficient through Eq. (1).

It is important to underline that the area A_{shell} defined above must not be confused with the total flow area available to the coolant of PCS. Indeed, the latter is given by:

$$A_{flow,tot,t,HX1} = N_{t,HX1} \cdot A_{flow,1tube,HX1} = 0.19 \text{ m}^2 \quad (4)$$

where $N_{t,HX1}$ is the total number of tubes in HX1 and $A_{flow,1tube,HX1}$ is the internal flow area associated with a single tube:

$$A_{flow,1tube,HX1} = \frac{\pi D_{in,t,HX1}^2}{4} \quad (5)$$

The pressure losses on the tube side can be evaluated using the well-known Colebrook or Haaland correlations for turbulent flow friction factors. In the specific case of U-tube heat exchangers, the tube curvature can be reasonably approximated as two consecutive 90° bends, each characterized by a local loss coefficient equal to:

$$K_{\text{loss}} = 0.45$$

Additional local pressure losses associated with sudden expansions and contractions can be estimated through the following correlations:

$$k_{\text{enl}} = \left(1 - \frac{A_1}{A_2}\right)^2 \quad (6)$$

$$k_{\text{contr}} = 0.5 \left(1 - \frac{A_2}{A_1}\right) \quad (7)$$

where A_1 represents the upstream flow area and A_2 the downstream flow area, namely the flow sections before and after the expansion or contraction. Equations (6) and (7) are employed to evaluate the local loss coefficients associated with the connections between pipes and headers, as well as between headers and tubes. Concerning the shell side, the overall local pressure loss coefficient can be estimated according to the following empirical correlation:

$$k_{\text{shell}} = 8 \left(\frac{0.227}{\text{Re}_{\text{shell}}^{0.193}} \right) \frac{D_{\text{shell}}}{D_{\text{eq,shell}}} (n_{\text{baffle}} + 1) \quad (8)$$

The evaluation of pressure losses will be incorporated into the subsequent calculations by considering which side of the heat exchanger is involved in each thermal-hydraulic loop. In particular, for the ISC, the shell-side pressure losses, represented by the coefficient k_{shell} , must be taken into account. Conversely, for the PSC, the dominant contribution is associated with the pressure losses occurring within the U-tubes on the tube side[4].

2.2 Heat Exchanger 2 (HX2):

HX2 is an in-pool type heat exchanger in which the ISC coolant flows through straight tubes immersed in a pool. Upstream and downstream manifolds are present, both characterized by the same cross-sectional area as the main ISC pipe.

In this case, the geometry specified in the problem formulation allows the analysis to be restricted exclusively to the ISC coolant circuit, namely the tube side of the heat exchanger. Consequently, the flow area associated with a single tube is:

$$A_{\text{flow,1tube,HX2}} = \frac{\pi D_{\text{in,t,HX2}}^2}{4} = 4.1259 \times 10^{-4} \text{ m}^2 \quad (9)$$

The total flow area available on the tube side is therefore:

$$A_{\text{flow,tot,t,HX2}} = N_{t,\text{HX2}} \cdot A_{\text{flow,1tube,HX2}} = 0.3177 \text{ m}^2 \quad (10)$$

Furthermore, in this configuration only the pressure losses associated with the ISC loop must be considered. These include distributed pressure losses due to pipe roughness, as well as concentrated losses occurring in the vicinity of the upstream and downstream manifolds.

3 Power Balance

It is important to evaluate the rate of energy generated in a reactor after shutdown for determining cooling requirements under normal conditions and accident consequences following abnormal events. In order to do so, it is instructive to first examine the decay energy sources for a reactor at power and then illustrate how these individual energy sources disappear or change upon shutdown. Let us consider a typical LWR reactor startup to 100% power, a period of operation and then a shutdown. The energy source components, neglecting the neutrino energy, are:

- Prompt fission power
- Fission product decay power (from ^{235}U , ^{238}U , ^{239}Pu and ^{241}Pu precursors)
- Delayed fission power
- Actinide decay power (from ^{241}Am , ^{241}Pu , ^{240}Pu , ^{239}Pu , ^{238}Pu , ^{244}Cm and ^{242}Cm). The contribution of ^{239}U and ^{239}Np decay is neglected [1]

In the present steady-state analysis, the full time-dependent decay heat curve is not tracked; instead, a conservative constant thermal power equal to 1% of the nominal reactor power is adopted. Consequently, the heat load that the DHRS must continuously remove is taken as $Q = 6 \text{ MW}$.

To simplify and clarify the analysis, the problem can be decomposed into three loops, each analogous to the one presented in "Introduction to Passive Decay Heat Removal Systems". Since the heat transfer process occurs under steady-state conditions, the thermal power generated in the reactor core must equal the heat transferred across each heat exchanger.

The energy balance for the Primary System Circuit (PSC) can be written as:

$$Q = \dot{m}_{PSC} c_{p,P} (T_{\text{hot},P} - T_{\text{cold},P}) \quad (11)$$

Similarly, the energy balance for the Intermediate System Circuit (ISC) is:

$$Q = \dot{m}_{ISC} c_{p,I} (T_{\text{hot},I} - T_{\text{cold},I}) \quad (12)$$

Furthermore, the Tertiary System Circuit (TSC), consisting of a pool at 1 bar under saturation conditions, is assumed to act as the ultimate heat sink of the system, thereby removing the entire thermal power.

4 Intermediate System Circuit

The analysis is conveniently initiated by considering the Intermediate System Circuit (ISC) in isolation. This choice is motivated by the fact that the Tertiary System Circuit (TSC) consists of a pool operating at atmospheric pressure under saturation conditions. As a result, its temperature is known a priori: $T_{\text{pool}} = 99.6^\circ\text{C}$.

Since the pool behaves as an effectively infinite thermal reservoir (ultimate heat sink), the temperatures within the ISC are constrained by this boundary condition. In contrast, the Primary System Circuit (PSC) does not have known temperatures a priori, as its thermal state is entirely dependent on the heat transfer processes occurring within the ISC.

The solution of the problem, given the available data, requires the adoption of an iterative approach. An initial estimate of the mass flow rate is assumed as $\dot{m}_{ISC} = 100 \text{ kg/s}$, together with an average temperature of 120°C . This temperature is used at each iteration to evaluate the thermophysical properties of the working fluid within the circuit by means of the XSteam library. At this stage, given the value of the mass flow rate in the duct and the specific isobaric heat capacity, Equation (2) can be used to determine the temperature rise ΔT_{ISC} required to remove the decay heat:

$$\Delta T_{ISC} = \frac{Q_{\text{decay}}}{\dot{m}_{ISC} c_p} \quad (13)$$

4.1 Energy Balance in Heat Exchanger 2

At this stage, an energy balance analysis on Heat Exchanger 2 (HX2) is required. This component represents the interface between the pool—where thermodynamic conditions are known—and the Intermediate Cooling System (ISC). Within HX2, heat is transferred to the pool, which is maintained under saturation conditions (i.e., 100°C at 1 bar).

The heat transfer mechanism is not governed by simple convection, but rather by boiling phenomena. The problem statement provides the following empirical correlation:

$$q'' = 2257 \Delta T_{\text{sat}} \left[\frac{\text{W}}{\text{m}^2} \right] \quad (14)$$

where q'' is in W/m^2 and ΔT_{sat} in K, so that the external boiling heat transfer coefficient is $h_{\text{ext}} = 2257 \text{ W/(m}^2\text{K)}$. which relates the heat flux q'' (in W/m^2) to the wall superheat, defined as:

$$\Delta T_{\text{sat}} = T_{\text{wall}} - T_{\text{sat}} \quad (15)$$

Since the total heat load ($Q = 6 \text{ MW}$) and the heat transfer area of HX2 (A_{HX2}) are known, the heat flux can be first evaluated as:

$$q'' = \frac{Q}{A_{\text{HX2}}} \quad (16)$$

By inverting the McAdams correlation, it is then possible to determine the required wall temperature, i.e., the temperature that the tube surface must reach in order to sustain the imposed heat transfer rate. Consequently, the external heat transfer coefficient on the pool side can be evaluated as:

$$h_{\text{ext}} = \frac{q''}{\Delta T_{\text{sat}}} \quad (17)$$

Only at this stage is it possible to evaluate the overall heat transfer transmittance U of the heat exchanger under consideration. In order to be transferred from the ISC fluid to the pool, heat must overcome three thermal resistances arranged in series:

- **Internal convection:** the ISC fluid flows inside the tubes (evaluated using the Dittus–Boelter correlation)³;
- **Wall conduction:** heat is conducted through the tube wall material;
- **External boiling:** heat is transferred from the outer tube surface to the pool (described by the McAdams correlation).

The overall heat transfer coefficient, referred to the external surface area, is therefore expressed as:

$$\frac{1}{U} = \frac{1}{h_{\text{ext}}} + \frac{D_{\text{out}} \ln \left(\frac{D_{\text{out}}}{D_{\text{in}}} \right)}{2 k_{\text{th}}} + \frac{D_{\text{out}}}{D_{\text{in}} h_{\text{int}}} \quad (18)$$

For a heat exchanger, the heat transfer rate is defined as:

$$Q = U \cdot A \cdot \Delta T_{\text{lm}} \quad (19)$$

where ΔT_{lm} is the logarithmic mean temperature difference between the two ends of the heat exchanger. Since the pool is maintained at a constant saturation temperature (T_{sat}), the expression simplifies to:

$$\Delta T_{\text{lm}} = \frac{(T_{\text{hot,ISC}} - T_{\text{sat}}) - (T_{\text{cold,ISC}} - T_{\text{sat}})}{\ln \left(\frac{T_{\text{hot,ISC}} - T_{\text{sat}}}{T_{\text{cold,ISC}} - T_{\text{sat}}} \right)} \quad (20)$$

In order to isolate the temperatures, the expression of the logarithmic mean temperature difference can be simplified. It is observed that the terms T_{sat} cancel out in the numerator, yielding $(T_{\text{hot}} - T_{\text{cold}})$, which is equal to ΔT_{ISC} . The equation can therefore be rewritten as:

$$\frac{Q}{U \cdot A} = \frac{\Delta T_{\text{ISC}}}{\ln \left(\frac{T_{\text{hot}} - T_{\text{sat}}}{T_{\text{cold}} - T_{\text{sat}}} \right)}$$

Rearranging the equation to isolate the logarithmic term:

³The internal convective heat transfer coefficient for the fluid flowing inside the heat exchanger can be evaluated using the relation $h_{\text{int}} = \frac{Nu_{\text{tube,HX2}} k_{\text{fluid,ISC}}}{D_{\text{in,t,HX2}}}$, where Nu_{HX2} denotes the Nusselt number for internal flow in the heat exchanger tubes, $k_{\text{fluid,ISC}}$ is the thermal conductivity of the working fluid, and $D_{\text{in,t,HX2}}$ is the internal diameter of the tubes. The Nusselt number is estimated using the Dittus–Boelter correlation for turbulent internal flow, given by $Nu = 0.023 Re^{0.8} Pr^n$, where Re is the Reynolds number, Pr is the Prandtl number, and the exponent n depends on the thermal condition of the fluid, taking the value $n = 0.4$ for heating and $n = 0.3$ for cooling.

$$\ln \left(\frac{T_{\text{hot}} - T_{\text{sat}}}{T_{\text{cold}} - T_{\text{sat}}} \right) = \frac{\Delta T_{\text{ISC}} \cdot U \cdot A}{Q}$$

Finally, by applying the exponential function to both sides, the temperatures can be expressed as:

$$\frac{T_{\text{hot}} - T_{\text{sat}}}{T_{\text{cold}} - T_{\text{sat}}} = \exp \left(\frac{\Delta T_{\text{ISC}} \cdot U \cdot A}{Q} \right)$$

It is known that:

$$T_{\text{hot}} = T_{\text{cold}} + \Delta T_{\text{ISC}} \quad (21)$$

Substituting this relation into the previous equation yields:

$$\frac{(T_{\text{cold}} + \Delta T_{\text{ISC}}) - T_{\text{sat}}}{T_{\text{cold}} - T_{\text{sat}}} = \exp \left(\frac{\Delta T_{\text{ISC}} \cdot U \cdot A}{Q} \right)$$

Solving this equation for the only remaining unknown, T_{cold} , leads to the final expression:

$$T_{\text{cold}} = T_{\text{sat}} + \frac{\Delta T_{\text{ISC}}}{\exp \left(\frac{\Delta T_{\text{ISC}} \cdot U \cdot A}{Q} \right) - 1}$$

Once T_{cold} is determined, the temperature of the hot branch can be directly obtained as:

$$T_{\text{hot}} = T_{\text{cold}} + \Delta T_{\text{ISC}}$$

4.2 Momentum Balance

Natural circulation arises from density differences induced by temperature gradients: hot water is less dense than cold water. The loop can be conceptually interpreted as a hydrostatic balance between two vertical branches of height H_{ISC} :

- In the hot leg, the fluid rises after being heated in HX1, with density ρ_{hot} ;
- In the cold leg, the fluid descends after being cooled in HX2, with density ρ_{cold} .

The resulting hydrostatic pressure difference between the two branches, referred to as the driving pressure ΔP_{D} , is given by:

$$\Delta P = g \cdot H_{\text{ISC}} \cdot (\rho_{\text{cold}} - \rho_{\text{hot}}) \quad (22)$$

The system reaches a steady-state condition when the buoyancy-driven pressure difference,

$$\Delta P_{\text{buoy}} = g \cdot H_{\text{ISC}} \cdot (\rho_{\text{cold}} - \rho_{\text{hot}}),$$

equals the total pressure losses within the hydraulic circuit:

$$\Delta P_{\text{buoy}} = \sum \Delta P_{\text{loss}}$$

where the total pressure losses, $\sum \Delta P_{\text{loss}}$, are composed of two main contributions: distributed losses and localized (minor) losses.

A. Distributed losses (pipe friction)

These losses arise from viscous friction between the fluid and the internal walls of straight pipes. They are typically evaluated using the Darcy–Weisbach equation:

$$\Delta P_{\text{dist}} = f \cdot \frac{L}{D} \cdot \frac{\rho v^2}{2}$$

where f is the Darcy friction factor, L is the pipe length, D is the pipe diameter, ρ is the fluid density, and v is the mean flow velocity.

B. Localized losses (minor losses)

These losses are associated with geometric discontinuities such as bends, contractions, expansions, valves, and fittings. They are commonly expressed as:

$$\Delta P_{loc} = k \cdot \frac{\rho v^2}{2}$$

Since the available data do not include the circuit mass flow rate nor the average temperature, the latter being required to evaluate the thermophysical properties of the coolant fluid (which, in this case, are obtained through the **XSteam** library implemented in MATLAB), the problem must be solved through an iterative procedure. Figure 3 reports the flowchart implemented in Matlab as a "while cycle" and used in the resolution of the iterative problem.

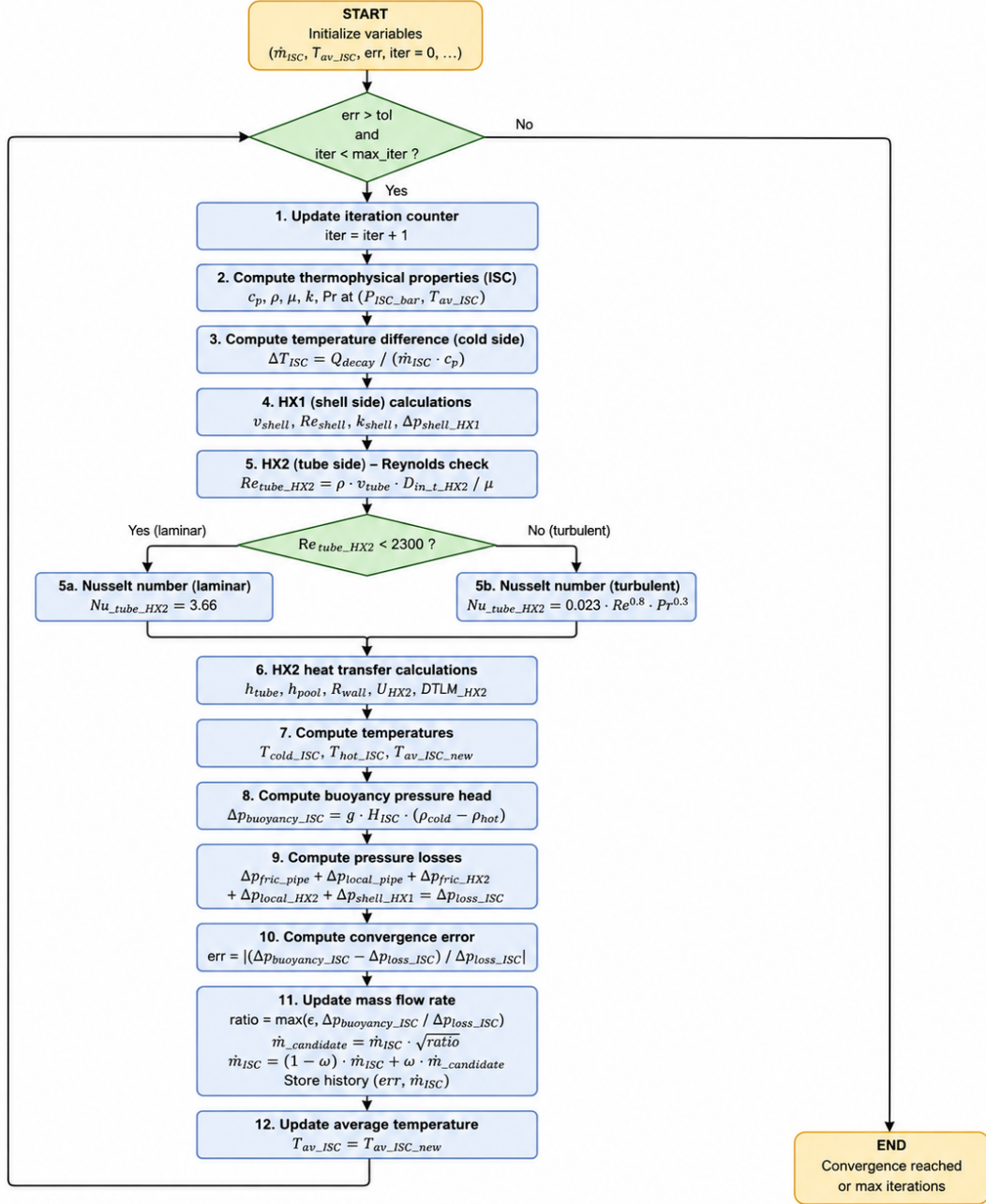


Figure 3: Iterative ISC thermal-hydraulic algorithm flowchart

5 Primary System Circuit (PSC)

The Primary System Circuit (PSC) is the fundamental cooling loop of the Decay Heat Removal System, directly responsible for extracting residual thermal power from the reactor core [?]. Its operation is based on purely passive principles, initiated by the opening of a check valve in the cold leg triggered by the pressure differential that arises when the primary coolant pumps stop [?]. Once circulation is established, the driving force is provided by the natural buoyancy resulting from the density gradient between the cold fluid in the vessel downcomer and the heated fluid within the core [?]. In the following analysis, the PSC is modeled at a steady-state condition where it must successfully exhaust 1% of the nominal thermal power, corresponding to 6 MW.

5.1 Thermal Coupling with HX1

Heat Exchanger 1 (HX1) serves as the thermal bridge between the Primary System Circuit and the Intermediate System Circuit (ISC). It is a shell-and-tube heat exchanger with U-tubes, where the primary coolant flows through the tube-side while the intermediate coolant flows on the shell-side in a countercurrent arrangement.

From a modeling perspective, the resolution of the PSC thermal state is coupled with the ISC. In the numerical algorithm developed for this analysis, the ISC parameters ($\dot{m}_{ISC}$, $T_{hot,ISC}$, and $T_{cold,ISC}$) are determined beforehand by anchoring the intermediate loop to the ultimate heat sink (the pool). Consequently, during the resolution of the PSC, the ISC acts as a thermal sink with known boundary conditions.

The heat transfer rate Q across HX1 must satisfy the global heat transfer equation:

$$Q = U_{HX1} \cdot A_{ht,HX1} \cdot F_T \cdot \Delta T_{ml,HX1} \quad (23)$$

Where:

- $Q = 6$ MW is the constant decay power to be removed [?];
- $A_{ht,HX1} = 500 \text{ m}^2$ is the total heat transfer area [?];
- $F_T = 0.7$ is the correction factor for the mean log temperature difference, accounting for the fact that HX1 is not a pure countercurrent exchanger;
- U_{HX1} is the overall heat transfer coefficient, calculated by summing the convective resistances (tube-side and shell-side) and the conductive resistance of the tube wall.

The logarithmic mean temperature difference ($\Delta T_{ml,HX1}$) for the countercurrent configuration is defined as:

$$\Delta T_{ml,HX1} = \frac{(T_{hot,PSC} - T_{hot,ISC}) - (T_{cold,PSC} - T_{cold,ISC})}{\ln \left(\frac{T_{hot,PSC} - T_{hot,ISC}}{T_{cold,PSC} - T_{cold,ISC}} \right)} \quad (24)$$

In the numerical solver, for each iteration of the mass flow rate $\dot{m}_{PSC}$, an internal loop finds the equilibrium temperatures that satisfy both the core energy balance:

$$Q = \dot{m}_{PSC} \cdot c_{p,PSC} \cdot (T_{hot,PSC} - T_{cold,PSC}) \quad (25)$$

and the heat exchanger performance defined in Eq. (1). The thermophysical properties of the water are evaluated at the constant PSC pressure of 75 bar. Due to the high efficiency of the countercurrent design, a "temperature cross" may be observed where the outlet temperature of the cold fluid (ISC) exceeds the outlet temperature of the hot fluid (PSC), a phenomenon correctly handled by the logarithmic mean formulation.

5.2 Momentum Balance and Core Integration

Once the thermal balance is established, the fluid properties—most importantly the density—can be evaluated. The natural circulation in the PSC is driven by the density difference between the cold downcomer and the hot core.

According to the system geometry, the net elevation is divided into two distinct segments: H_1 , the net elevation difference between the highest points of the hot and cold legs, and H_2 , the elevation difference between the top and the bottom of the core.

In the H_1 region, the fluid temperature and density are assumed to be constant in both the rising hot leg and the descending cold leg. The corresponding buoyancy contribution is:

$$\Delta P_{\text{buoy},H1} = g \cdot H_1 \cdot (\rho_{\text{cold}} - \rho_{\text{hot}}) \quad (26)$$

Within the core region (H_2), the fluid is progressively heated as it flows upward. Assuming a linear density variation along the core height, the average density of the fluid inside the core is the arithmetic mean of the inlet (ρ_{cold}) and outlet (ρ_{hot}) densities. The resulting buoyancy contribution from the core is the difference between the hydrostatic pressure of the cold downcomer column and the hot core column:

$$\Delta P_{\text{buoy},H2} = \rho_{\text{cold}} g H_2 - \left(\frac{\rho_{\text{cold}} + \rho_{\text{hot}}}{2} \right) g H_2 = g \cdot H_2 \cdot \frac{\rho_{\text{cold}} - \rho_{\text{hot}}}{2} \quad (27)$$

The total driving pressure for the Primary System Circuit is the sum of these two contributions: $\Delta P_{\text{buoy},\text{PSC}} = \Delta P_{\text{buoy},H1} + \Delta P_{\text{buoy},H2}$.

5.3 Vessel and Loop Pressure Drops

In steady-state conditions, the driving force is perfectly balanced by the total pressure drops along the circuit. These include distributed friction in the main pipe and HX1 U-tubes (evaluated via the Haaland equation), localized losses due to bends, the check valve, and the transitions at the HX1 headers.

A specific treatment is required for the pressure drop across the reactor vessel. It is modeled by assuming a constant concentrated loss coefficient K_{vessel} , valid under the hypothesis that the flow pattern does not change dramatically with respect to the nominal condition. From the nominal data ($\Delta p_{\text{vessel}} = 1.2$ bar at $\dot{m} = 3200$ kg/s) one obtains:

$$K_{\text{vessel}} = \frac{\Delta p_{\text{vessel}}}{\dot{m}^2} = \frac{1.2 \times 10^5 \text{ Pa}}{(3200 \text{ kg/s})^2} \quad (28)$$

During natural circulation, the vessel pressure loss is then evaluated dynamically as $\Delta P_{\text{vessel}} = K_{\text{vessel}} \cdot \dot{m}_{\text{PSC}}^2$.

6 Numerical Methodology

Due to the highly non-linear nature of the governing equations—where heat transfer coefficients depend on mass flow rates, and mass flow rates depend on temperatures via buoyancy—an iterative numerical algorithm was developed in MATLAB.

6.1 Iterative Scheme

To avoid multi-dimensional instabilities, the system of equations is decoupled by sequentially solving the ISC and the PSC. The pool in the TSC provides a fixed temperature boundary condition (T_{sat}) that anchors the entire calculation.

The algorithm begins by solving the isolated ISC loop. For each iteration:

1. Thermophysical properties are evaluated at the assumed average temperature and constant pressure using the `XSteam` library[3].
2. The thermal balance is solved to determine the hot and cold temperatures. For HX2, the ε -NTU method is implemented to ensure numerical stability and avoid undefined logarithms when the fluid temperatures approach the pool saturation temperature.
3. The momentum balance is evaluated by computing the buoyancy force and the total pressure losses, explicitly accounting for localized velocity variations at each geometric transition.

Once the ISC converges, its temperatures and mass flow rate are used as fixed boundary conditions for the shell-side of HX1, allowing the PSC to be solved using a similar nested iterative approach.

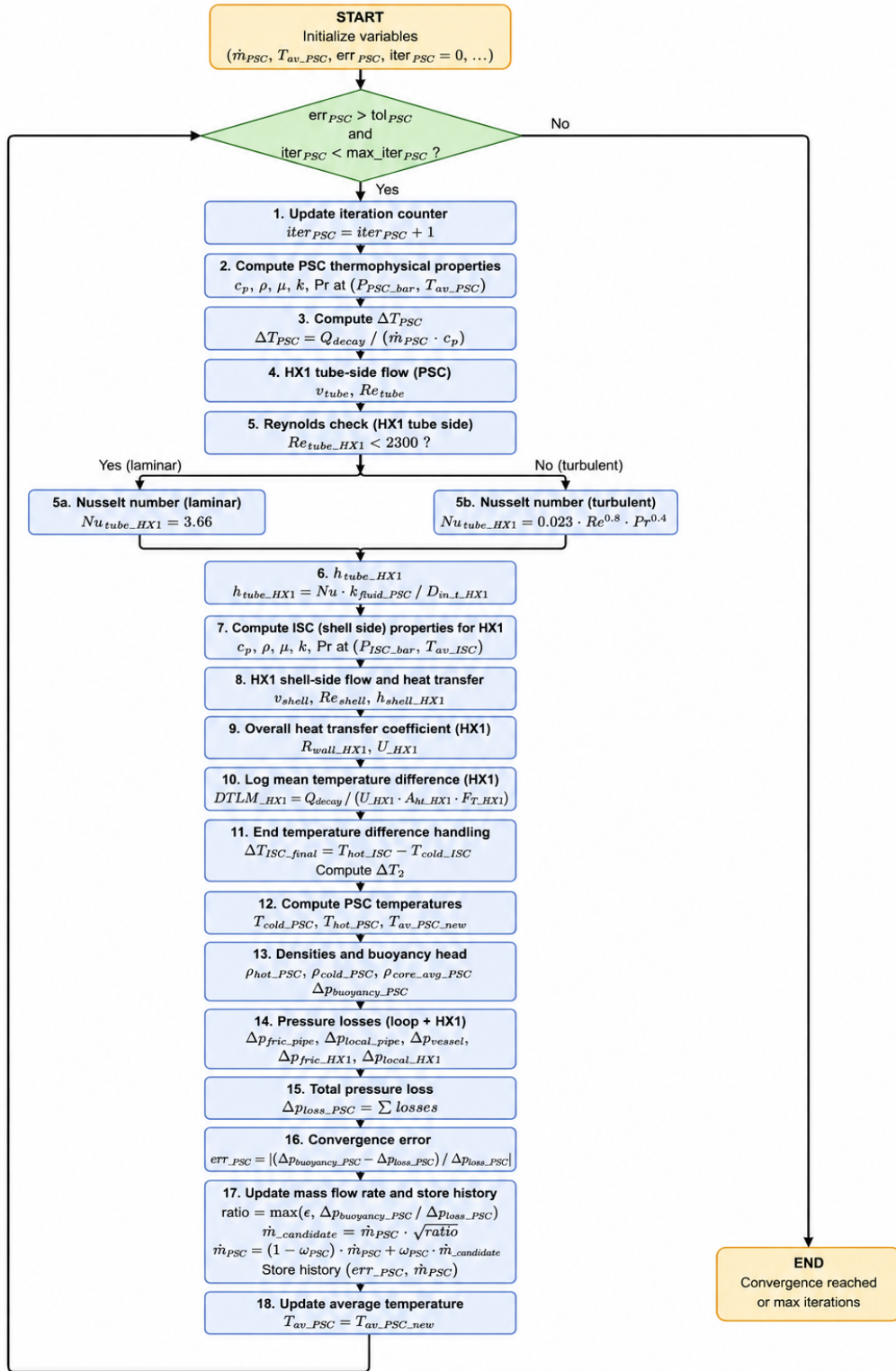


Figure 4: Iterative PSC thermal-hydraulic algorithm flowchart

6.2 Convergence and Relaxation

If the calculated buoyancy force does not match the pressure losses, the mass flow rate must be updated. Recognizing that fluid dynamic pressure drops are strictly proportional to the square of the mass flow rate ($\Delta P_{\text{loss}} \propto \dot{m}^2$), a robust relaxation method based on the square root of the pressure ratio was implemented:

$$\dot{m}_{\text{new}} = \dot{m}_{\text{old}} \sqrt{\frac{\Delta P_{\text{buoyancy}}}{\Delta P_{\text{loss}}}} \quad (29)$$

This physically-based updating scheme significantly accelerates convergence, preventing numerical oscillations and typically reaching the predefined relative tolerance of 10^{-4} in fewer than 20 iterations. To further guarantee stability, absolute value constraints are applied to the pressure ratio to prevent the evaluation of complex numbers during initial unstable iterations. The stability and efficiency of the implemented relaxation scheme are visually confirmed by the convergence plots. As shown in Figure 5, the relative residuals for both the ISC and PSC drop monotonically and exponentially, reaching the required tolerance in roughly 12-13 iterations. Correspondingly, Figure 6 illustrates how the mass flow rates quickly stabilize to their steady-state values without any numerical oscillations.

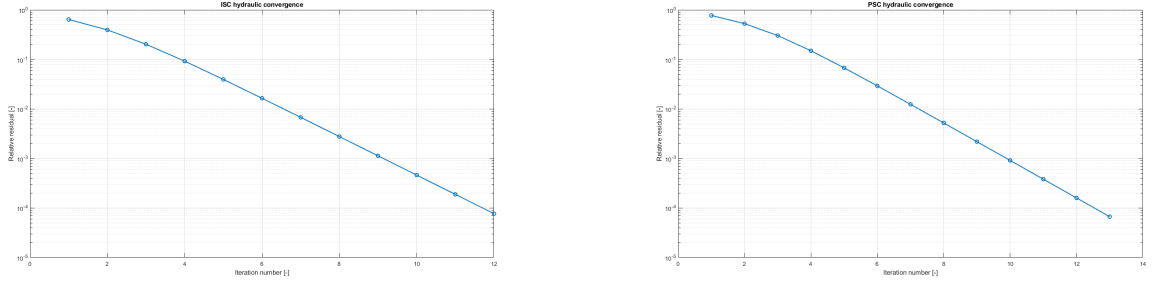


Figure 5: Hydraulic convergence (relative residual) for the ISC (left) and PSC (right).

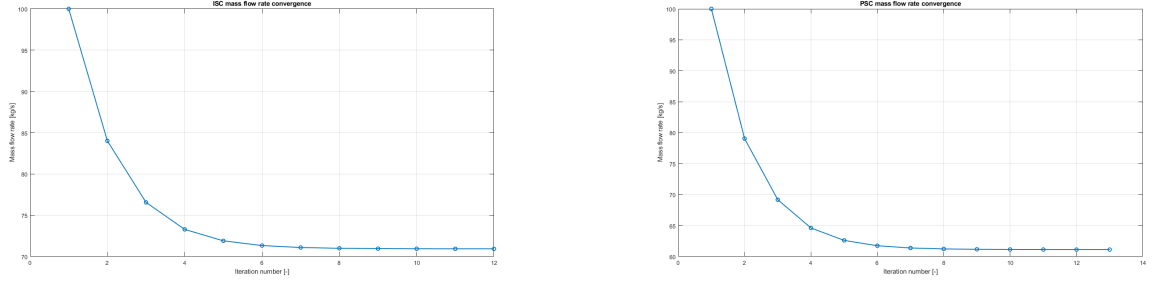


Figure 6: Mass flow rate convergence for the ISC (left) and PSC (right).

7 Results and Discussion

The implementation of the numerical algorithm allowed the evaluation of the steady-state operating conditions of the DHRS. The results successfully demonstrate that natural circulation alone is capable of establishing a stable mass flow rate sufficient to exhaust the 6 MW of decay heat.

7.1 Intermediate System Circuit Results

The Intermediate System Circuit reached convergence in 12 iterations. The steady-state mass flow rate was found to be $\dot{m}_{ISC} = 70.935$ kg/s. The fluid temperatures at the boundaries of the circuit resulted in $T_{\text{hot},ISC} = 129.08^\circ\text{C}$ and $T_{\text{cold},ISC} = 109.07^\circ\text{C}$.

The total driving force (buoyancy) evaluated for the ISC is 1556.37 Pa. A detailed breakdown of the pressure losses perfectly balancing this driving force is presented in Table 5. As expected for a loop characterized by a significant length ($L = 40$ m), the dominant losses are the distributed friction and localized losses along the main pipe, alongside the concentrated losses in HX2.

Table 1: Pressure drop distribution in the Intermediate System Circuit (ISC)

ISC Component	Pressure Drop [Pa]
Main Pipe Friction	450.39
Main Pipe Local Losses	246.71
HX1 Shell-Side Pressure Loss	204.98
HX2 Tube-Side Friction	204.10
HX2 Local Losses	450.32
TOTAL LOSSES	1556.49
BUOYANCY	1556.37

A visual representation of the pressure drop breakdown is provided in Figure 7. The bar chart clearly highlights the dominance of the main pipe friction and the localized losses in HX2, visually confirming the data reported in the table.

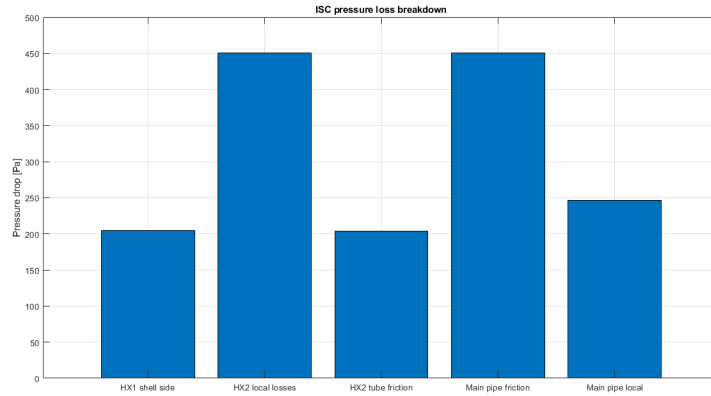


Figure 7: Breakdown of the pressure losses within the Intermediate System Circuit.

7.2 Primary System Circuit Results

The Primary System Circuit solver achieved convergence in 13 iterations, yielding a steady-state mass flow rate of $\dot{m}_{PSC} = 61.125$ kg/s. The core outlet temperature (hot leg) stabilized at $T_{hot,PSC} = 148.47^\circ\text{C}$, while the core inlet temperature (cold leg) resulted in $T_{cold,PSC} = 125.42^\circ\text{C}$.

The buoyancy force driving the primary loop is 1666.58 Pa. The breakdown of the corresponding pressure drops is detailed in Table 6. Unlike the ISC, the PSC is heavily penalized by the friction within the 897 U-tubes of HX1, which accounts for nearly half of the total pressure losses (733.34 Pa). The reactor vessel, at these low natural circulation flow rates, opposes a minimal resistance (43.79 Pa) compared to nominal active conditions.

Table 2: Pressure drop distribution in the Primary System Circuit (PSC)

PSC Component	Pressure Drop [Pa]
Reactor Vessel	43.79
Main Pipe Friction	135.86
Main Pipe Local Losses (Bends & Valve)	396.92
HX1 Tube-Side Friction	733.34
HX1 Local Losses (U-bends & Headers)	356.78
TOTAL LOSSES	1666.69
BUOYANCY	1666.58

The distribution of these pressure losses is illustrated in the bar chart of Figure 8. The graphical representation makes it immediately evident how the friction within the narrow U-tubes of HX1 constitutes the most severe hydraulic penalty in the primary circuit, vastly exceeding the resistance opposed by the reactor vessel.

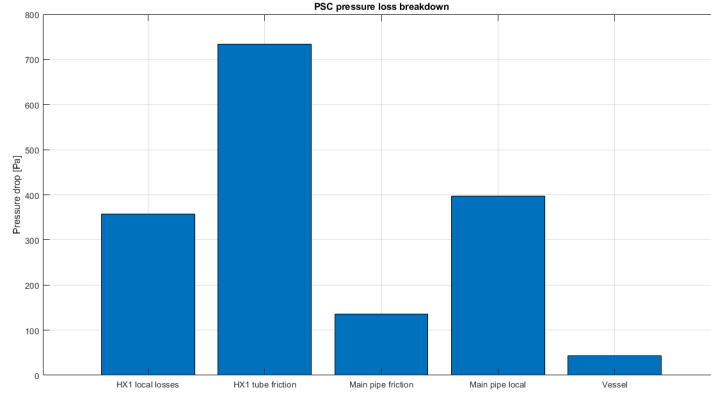


Figure 8: Breakdown of the pressure losses within the Primary System Circuit.

7.3 Physical Interpretation of the Results

The calculated results exhibit strict adherence to thermodynamic and fluid dynamic principles:

- **Thermal Cascade:** The fundamental requirement for passive heat removal is the spontaneous flow of heat from the source to the ultimate sink. The temperatures rigorously follow this downward cascade: the core peak temperature (148.47°C) is higher than the ISC peak temperature (129.08°C), which in turn is strictly higher than the pool saturation temperature ($\approx 100^{\circ}\text{C}$), guaranteeing a continuous positive ΔT across all heat exchangers.
- **Temperature Cross in HX1:** A notable feature of the results is that the hot outlet temperature of the ISC (129.08°C) is higher than the cold outlet temperature of the PSC (125.42°C). While this might appear counter-intuitive, it is the hallmark of a highly efficient countercurrent heat exchanger. The fluids flow in opposite directions, allowing the exiting cold fluid to exchange heat with the entering hot fluid at its maximum temperature peak, thereby achieving a physical "temperature cross" impossible in co-current setups.
- **Mass Flow Rate Behavior:** The mass flow rate in the PSC ($\approx 61.1 \text{ kg/s}$) is lower than that in the ISC ($\approx 70.9 \text{ kg/s}$). This phenomenon is driven by the severe hydraulic resistance imposed by the narrow U-tubes of HX1 on the primary side. Since $\Delta P_{\text{loss}} \propto \dot{m}^2$, a circuit with intrinsically higher loss coefficients will inherently balance the available buoyancy at a lower steady-state velocity.

Conclusions

The present work investigated the steady-state thermal–hydraulic behavior of a passive Decay Heat Removal System (DHRS) for a nuclear reactor operating under post-shutdown conditions. By assuming a residual decay heat equal to 1% of the nominal reactor thermal power, corresponding to 6 MW, it was possible to evaluate whether natural circulation alone could ensure adequate heat removal without the support of active components such as pumps.

The analysis demonstrated that the system is capable of establishing stable natural circulation in both the Primary System Circuit (PSC) and the Intermediate System Circuit (ISC). The calculated mass flow rates and temperature distributions were found to be physically consistent and sufficient to guarantee continuous heat transfer from the reactor core to the ultimate heat sink represented by the atmospheric pool.

Particular attention was devoted to the thermal coupling between the circuits through Heat Exchangers HX1 and HX2. The adopted methodology highlighted the importance of accurately accounting for both distributed and localized pressure losses, especially within the U-tube geometry of HX1, which was shown to represent the dominant hydraulic resistance of the PSC. At the same time, the shell-side behavior of HX1 and the boiling heat transfer occurring in HX2 played a fundamental role in determining the global thermal performance of the system.

From a numerical standpoint, the iterative algorithm implemented in MATLAB proved to be stable and computationally efficient. The use of relaxation techniques based on the square root of the pressure ratio allowed rapid convergence while avoiding oscillatory behavior during the solution process. Furthermore, the integration of the **XSteam** library enabled a realistic evaluation of the thermophysical properties of the working fluid as a function of temperature and pressure.

Beyond the numerical results themselves, the study provides a clear physical interpretation of passive safety systems. The entire DHRS operation is governed by simple but robust physical principles: density differences induced by temperature gradients naturally generate the driving force required for circulation and heat removal. This characteristic represents one of the major advantages of passive systems in nuclear engineering, as their functionality does not rely on external power supplies or mechanically driven components.

Overall, the obtained results confirm the effectiveness of the analyzed passive DHRS configuration in maintaining safe thermal conditions after reactor shutdown. Although the model relies on simplifying assumptions—such as steady-state conditions, constant pressure levels, and the neglect of fouling resistances—the developed approach captures the essential thermal–hydraulic mechanisms governing the system behavior and provides a solid basis for more advanced transient or multi-physics analyses.

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